
STATE OF AI RESEARCH

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ABSTRACT

This study analyzes over five million academic works that discuss artificial intelligence methods in their abstracts, published between 2013 and mid-2026. The analysis measures keyword frequency, temporal trajectories, growth rates, citation distributions, geographic concentration, institutional output, open access rates, and title-versus-abstract coverage gaps. Key findings include the continued dominance of neural networks across 30% of all abstracts, the 29.9x growth of large language model papers from 2018 to 2025, China exceeding the United States in total AI research volume by paper count, and nearly half of all papers receiving zero citations.

1 INTRODUCTION

In 2025, over 944,000 academic works in the OpenAlex database mentioned artificial intelligence methods in their abstracts. Through early June 2026, 812,972 such papers have been recorded, putting the full year on pace for approximately 1.6 million total papers.

Many bibliometric studies of AI research rely on subject classification tags or curated keyword lists that may not capture cross-disciplinary usage of AI methods. When keyword search is used, title-based approaches capture only papers where the author chose to place the method name in the title. A paper titled “Predicting Protein Stability Under Thermal Stress” that uses a neural network throughout its methods section would be invisible to a title-only search for “neural network.” Abstract-level analysis addresses this gap by searching the text where authors describe their methods, results, and contributions.

To analyze these trends, a bibliometric corpus of 5,003,783 publications was defined by querying the OpenAlex scholarly database for academic works published between 2013 and mid-2026 that explicitly mention AI-related terms in their abstracts. No papers were downloaded; all analyses were performed through API count queries.

The analysis covers 14 annual cohorts (2013-2026), measuring publication volume, n-gram frequency, growth rates, citation distributions, geographic output, and open access rates, and compares abstract-level search against title-only search to quantify the coverage gap.

The rest of this paper is organized as follows. Section 2 describes the corpus definition and analysis methods. Section 3 presents results across nine dimensions. Section 4 discusses the findings. Section 5 reviews related work. Section 6 concludes.

2 METHODOLOGY

2.1 DATA SOURCE

OpenAlex is an open scholarly database indexing over 250 million academic works [1; 19]. It provides free API access with structured metadata including titles, abstracts (stored as inverted index), author affiliations, citation counts, open access status, and machine-learned concept tags. OpenAlex was chosen over Web of Science or Scopus because it is freely accessible, covers preprints (including arXiv), and exposes structured API filters that allow exact-count queries without downloading raw data.

2.2 CORPUS DEFINITION

The corpus was defined using OpenAlex’s `abstract.search` filter. The search query combined 10 AI-related terms using boolean OR logic.

Search terms. *artificial intelligence, machine learning, deep learning, neural network, language model, reinforcement learning, computer vision, natural language, generative, autonomous.*

Date range. 2013-01-01 through 2026-06-08.

Resulting corpus. 5,003,783 papers.

Eight of the ten search terms are specific to AI. Two terms (“generative” and “autonomous”) are broader and match non-AI papers. “Generative” can match generative grammar in linguistics. “Autonomous” can match autonomous systems in biology. Some fraction of the corpus consists of papers where the matched term refers to a non-AI context. This tradeoff was accepted because excluding these terms would miss entire AI research areas such as generative adversarial networks [10] (67,647 papers) and autonomous driving (71,930 papers).

Table 1: Corpus composition by document type and source type.

Document Type	Count	Share	Source Type	Count	Share
Journal article	3,270,717	65.4%	Journal	2,232,178	44.6%
Preprint	668,948	13.4%	Repository (arXiv, SSRN, etc.)	1,370,590	27.4%
Book chapter	386,449	7.7%	Book series	246,842	4.9%
Dataset	298,570	6.0%	Conference proceedings	111,398	2.2%
Review	94,567	1.9%	eBook platform	69,676	1.4%
Dissertation	92,581	1.9%	Other / unclassified	973,099	19.5%
Other	191,951	3.8%			
Total	5,003,783				

Repositories (primarily arXiv) account for 27.4% of the corpus, reflecting the significant role of preprints in AI research dissemination. Journals remain the largest single source at 44.6%.

2.3 ANALYSIS METHODS

All analyses were performed through direct OpenAlex API calls. No local text processing was applied.

1. **Keyword frequency.** For each keyword, bigram, or trigram of interest, a single API call was issued using `abstract.search:<term>,publication_year:2013-2026` and the `meta.count` field from the response was recorded.
2. **Growth detection.** For each keyword, the total publication count from 2025-2026 was divided by the total from 2022-2023 to produce a growth ratio.
3. **Time-series trajectories.** For 10 selected methods, year-by-year API calls were issued to obtain annual publication counts from 2013 (or the method’s introduction year) through 2026.
4. **Citation distribution.** The `cited_by_count` filter was used to count papers in seven citation ranges (0, 1-10, 11-50, 51-100, 101-500, 501-1000, 1000+).
5. **Geographic and institutional analysis.** OpenAlex’s `group_by` aggregation on `authorships.countries` and `authorships.institutions.lineage` was used to rank countries and institutions. A single paper with co-authors from multiple countries is counted once per country.
6. **Title vs. abstract comparison.** For 12 keywords, parallel queries were run using `title.search` and `abstract.search` and the ratio of abstract to title counts was computed.

2.4 STEMMING AND PRECISION

OpenAlex’s search filters apply stemming, meaning a search for “agentic” also matches “agent” and “agents.” For multi-word phrases (“retrieval augmented generation,” “graph neural network”) and proper nouns (“DeepSeek,” “Claude,” “Mistral”), stemming has minimal effect. For single common words (“diffusion,” “safety,” “clinical”), stemming can inflate counts by matching non-AI uses of the word. This limitation is discussed further in Section 4.5.

3 RESULTS

3.1 PUBLICATION VOLUME

Table 2: Annual publication volume (abstract-level corpus).

Year	Papers (Count)	YoY Growth	Cumulative Count	Share of Corpus
2013	93,226	-	93,226	1.9%
2014	97,510	+4.6%	190,736	1.9%
2015	105,609	+8.3%	296,345	2.1%
2016	115,423	+9.3%	411,768	2.3%
2017	137,237	+18.9%	549,005	2.7%
2018	185,192	+34.9%	734,197	3.7%
2019	242,286	+30.8%	976,483	4.8%
2020	305,903	+26.3%	1,282,386	6.1%
2021	369,519	+20.8%	1,651,905	7.4%
2022	411,098	+11.2%	2,063,003	8.2%
2023	520,861	+26.7%	2,583,864	10.4%
2024	662,417	+27.2%	3,246,281	13.2%
2025	944,530	+42.6%	4,190,811	18.9%
2026 (up to June)	812,972	-	5,003,783	16.2%

The corpus grew from 93,226 papers in 2013 to 944,530 in 2025, a 10.1x increase over 12 years. Part of this growth reflects the general expansion of academic publishing, but AI growth has consistently outpaced the baseline, particularly after 2017. The growth curve shows three distinct phases.

Phase 1 (2013-2016), slow growth, 4.6-9.3% per year. AI research was growing but had not yet reached mainstream adoption. Deep learning was still an active research area rather than a standard tool.

Phase 2 (2017-2022), deep learning adoption, 11.2-34.9% per year. Growth was fastest in 2017-2018 (+18.9% and +34.9%), aligning with the publication of “Attention Is All You Need” [6] and the broad adoption of deep learning across application domains. Growth decelerated to 11.2% in 2022.

Phase 3 (2023-2026), the LLM surge, 26.7-42.6% per year. Starting in 2023, growth re-accelerated sharply. The 2025 output (944,530) represents a 42.6% increase over 2024, the highest annual growth rate since 2018. This aligns with the release of ChatGPT (November 2022) and the subsequent expansion of LLM-related research.

The 2026 cohort (812,972 papers recorded through early June) is on pace to reach approximately 1.6 million publications for the full year based on linear extrapolation. If realized, this would be the first year in which annual AI research output exceeds 1 million papers in the corpus.

For context, OpenAlex indexes over 250 million works across all fields and time periods. The AI corpus of 5 million works spans 2013 through mid-2026, a 13.5-year window. Within that window, annual AI output grew from 93,226 works in 2013 to 944,530 in 2025, a 10.1x increase.

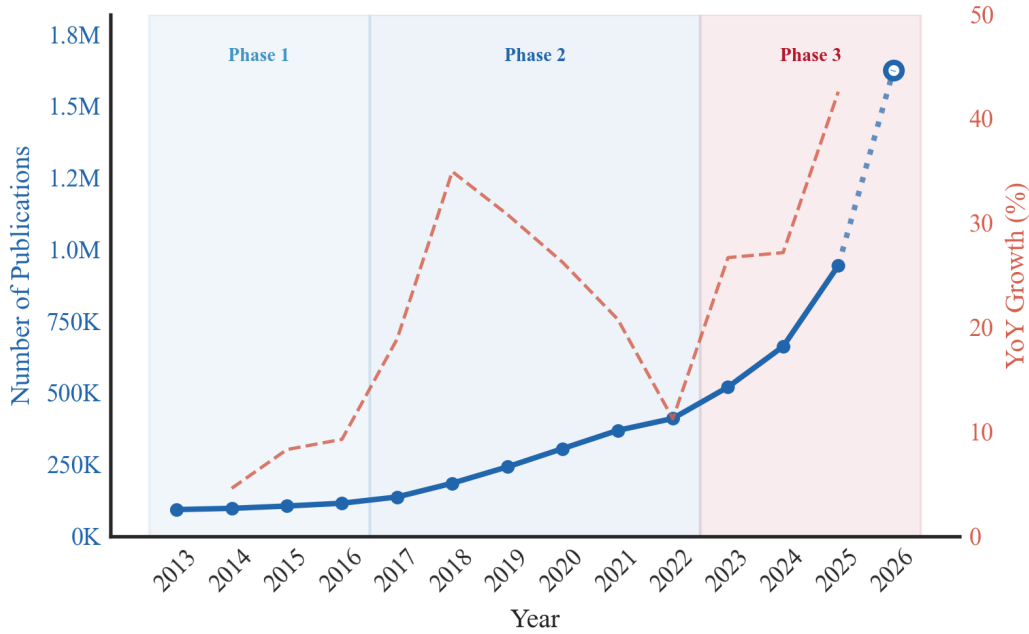


Figure 1: AI research publication volume, 2013-2026. The 2026 value is an annualized estimate based on January-June data (open marker).

3.2 N-GRAM FREQUENCY

Table 3: Top 10 bigrams and trigrams in abstracts.

Rank	Bigram	Mentions	Trigram	Mentions
1	neural network	1,522,612	deep neural network	518,431
2	machine learning	1,287,123	convolutional neural network	394,934
3	deep learning	980,070	large language model	292,873
4	artificial intelligence	745,358	artificial neural network	261,355
5	attention mechanism	432,079	support vector machine	239,347
6	large language	405,166	natural language processing	172,355
7	image classification	390,138	long short-term memory	137,359
8	recommendation system	387,638	recurrent neural network	88,266
9	medical imaging	359,104	graph neural network	86,453
10	feature extraction	256,159	random forest classifier	73,385

“Neural network” (1,522,612) dominates the list, appearing in 30.4% of all paper abstracts. “Machine learning” (1,287,123) and “deep learning” (980,070) round out the top three. Together, these three terms account for over 3.8 million abstract mentions in total (including duplicate counts where multiple terms appear in the same abstract).

“Attention mechanism” (432,079) ranks 5th, reflecting the wide adoption of attention-based architectures across NLP, computer vision, and multimodal tasks since the transformer’s introduction in 2017.

“Large language” (405,166) at rank 6 captures the LLM wave. It already surpasses older application-oriented terms such as “image classification” (390,138), “recommendation system” (387,638), and “medical imaging” (359,104), demonstrating how quickly the LLM category has accumulated volume.

“Feature extraction” (256,159) at rank 10 reflects how many AI papers describe preprocessing pipelines rather than end-to-end architectures.

“Deep neural network” (518,431) leads the trigrams, followed by “convolutional neural network” (394,934) [3]. Together these account for over 913,000 abstract mentions, reflecting the dominance of image-related deep learning tasks in the corpus.

“Large language model” (292,873) at rank 3 has overtaken “artificial neural network” (261,355) and “support vector machine” (239,347). The abstract data reveals that LLMs are discussed more broadly than title-only searches suggest (292,873 abstract mentions vs. 71,469 title mentions, a 4.1x ratio per Table 6).

“Support vector machine” (239,347) and “random forest classifier” (73,385) persist in the top 10. These classical methods continue to be widely referenced in abstracts, often as baselines for comparison or in application domains where simpler models remain competitive.

3.3 TIME-SERIES TRAJECTORIES

Foundational methods. “Neural network” grew steadily from 23,395 abstract mentions in 2013 to 207,140 in 2025 (representing an 8.9x increase). “Deep learning” grew from 4,120 in 2013 to 216,713 in 2025 (a 52.6x increase), narrowing the gap with “neural network” though not yet surpassing it in annual counts. “Reinforcement learning” grew from 1,784 in 2013 to 47,498 in 2025 (a 26.6x increase). “Transformer” grew from 7,201 abstract mentions in 2017 to 78,135 in 2025 (a 10.9x increase). The early counts (2017-2018) likely include non-AI uses of “transformer” (e.g., electrical engineering).

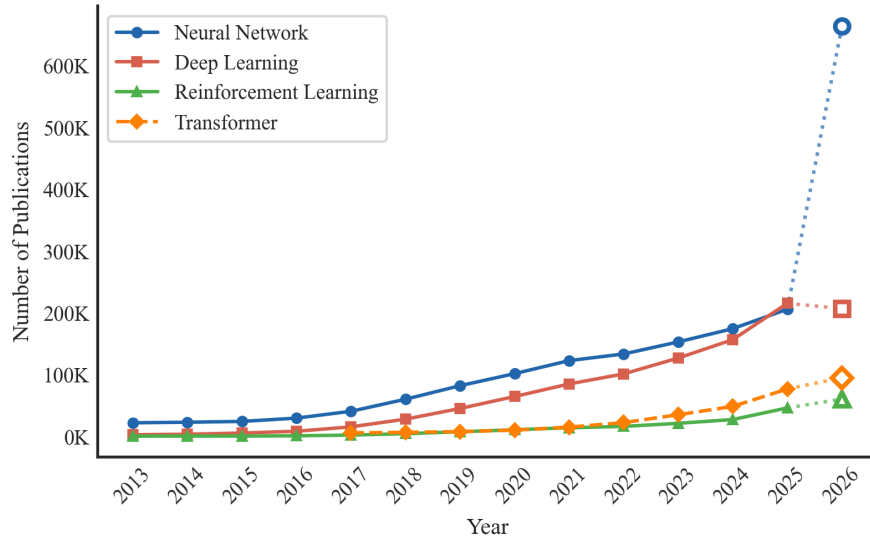


Figure 2: Trajectories of foundational AI methods by annual abstract mentions, 2013-2026. The 2026 values are annualized estimates based on January-June data (open markers).

The LLM trajectory. “Large language model” abstract mentions grew from 3,248 in 2018 to 96,984 in 2025, representing a 29.9x increase. The growth curve has a clear inflection point. Between 2018 and 2022, mentions grew at a modest pace (from 3,248 to 7,931 mentions, or a 2.4x increase over four years). Between 2022 and 2025, mentions grew 12.2x in three years. The annualized 2026 estimate (approximately 170,000 papers, based on 84,957 recorded through June) suggests another 1.8x increase over the 2025 volume.

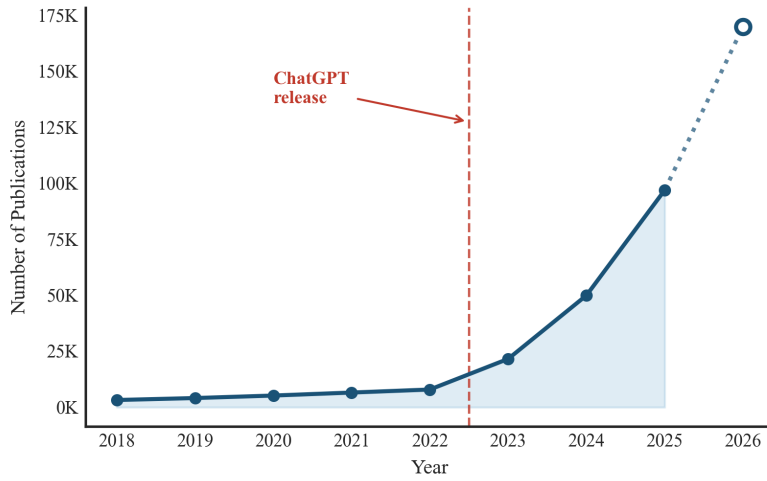


Figure 3: Growth of "large language model" in paper abstracts. The 2026 value is an annualized estimate (open marker).

Rising methods. "Diffusion model" grew from 18,640 abstract mentions in 2019 to 49,862 in 2025 (a 2.7x increase). Early counts include non-AI uses of "diffusion model" (e.g., diffusion of innovations in social science). "Federated learning" [12] grew from 46 mentions in 2017 to 18,519 in 2025 (a 402.6x increase). "Graph neural" grew from 966 mentions in 2017 to 21,873 in 2025 (a 22.6x increase). "Knowledge graph" grew from 1,700 mentions in 2013 to 16,519 in 2025 (a 9.7x increase).

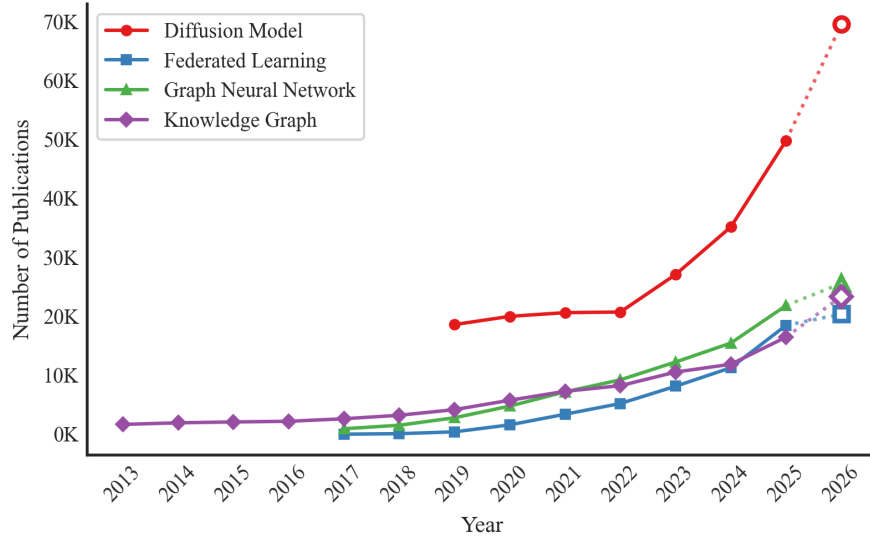


Figure 4: Rising AI methods by annual abstract mentions. The 2026 values are annualized estimates based on January-June data (open markers).

"Generative adversarial" grew from 6 papers in 2014 to 13,613 in 2025. Its slowing growth is discussed in Section 4.3.

3.4 FASTEST-RISING KEYWORDS

Table 4: Top 10 fastest-rising keywords in paper abstracts (2025-2026 vs. 2022-2023).

Keyword	2025-2026 (Count)	2022-2023 (Count)	Growth (Ratio)
deepseek	11,033	13	848.7x
retrieval augmented generation	18,196	347	52.4x
jailbreak	2,803	110	25.5x
retrieval-augmented	21,105	1,101	19.2x
mistral	4,361	260	16.8x
llm	161,771	10,125	16.0x
copilot	5,699	356	16.0x
rag	19,193	1,250	15.4x
gemini	22,365	1,650	13.6x
guardrail	5,046	521	9.7x

The keyword growth data highlights three primary research trends.

Story 1, the model name explosion. “DeepSeek” (848.7x), “Mistral” (16.8x), and “Gemini” (13.6x) are all names of specific models. Researchers are studying specific products, not just abstract architectures. The field is increasingly focused on model-level evaluation and comparison alongside architecture research.

Story 2, the RAG pipeline. “Retrieval augmented generation” (52.4x), “retrieval-augmented” (19.2x), and “RAG” (15.4x) all show rapid growth. RAG has become the standard pattern for connecting language models to external knowledge bases [8]. Its three variants in the growth table reflect how quickly researchers adopted it as both a technique and an abbreviation.

Story 3, safety and reliability. “Jailbreak” (25.5x) and “guardrail” (9.7x) reflect the growing research effort to make language models reliable and safe. “Jailbreak” research (2,803 papers) investigates adversarial prompts that circumvent model safety filters. “Guardrail” (5,046) covers techniques for constraining model outputs. These terms barely existed in the research literature before 2023.

3.5 CITATION DISTRIBUTION

The citation distribution is extremely right-skewed across the corpus. Nearly half of all papers (48.9%) have received zero citations to date. Only 2,475 papers (0.05%) have accumulated more than 1,000 citations. Consequently, the median paper in this AI corpus has zero citations. This figure is partly inflated by recency: papers published in 2024-2026 have had little time to accumulate citations.

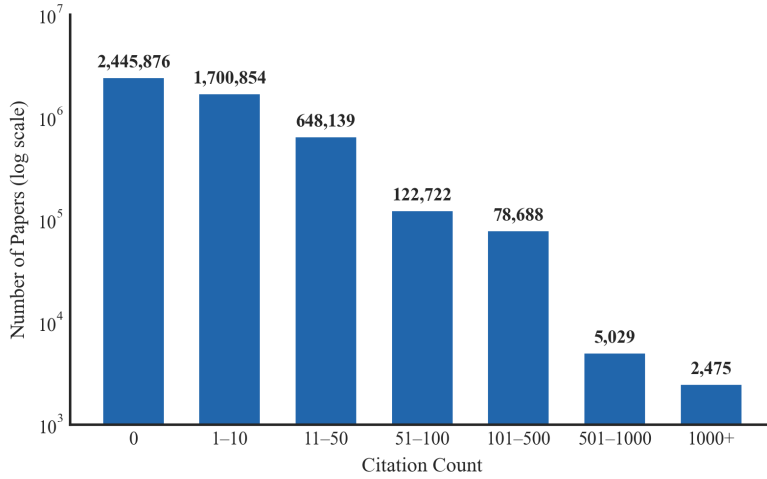


Figure 5: Citation distribution of AI papers on a log scale. Nearly half of all papers have zero citations. Only 2,475 papers have exceeded 1,000 citations.

The most-cited paper is ResNet [2] with 221,202 citations, approximately 1.9x the next entry. Several of the most-cited papers in the corpus, including the DSM-5 [23] (113,579 citations) and the lme4 statistics package [24] (84,949), are not research contributions to the field but mention related methods in their abstracts, illustrating the breadth of abstract-level search. Among field-specific papers, the top entries (ResNet, Deep Learning [5], AlexNet, VGGNet, Faster R-CNN, XGBoost) are all foundational infrastructure. Papers that provide widely-used building blocks receive orders of magnitude more citations than application-specific work.

3.6 GEOGRAPHIC DISTRIBUTION

In terms of total geographic distribution of research output in the corpus, authors affiliated with Chinese institutions lead with 874,019 publications, which is 21.6% higher than the output of authors affiliated with US institutions (718,676). India ranks third with 369,931 publications, surpassing the outputs of Japan (333,896) and the United Kingdom (216,177).

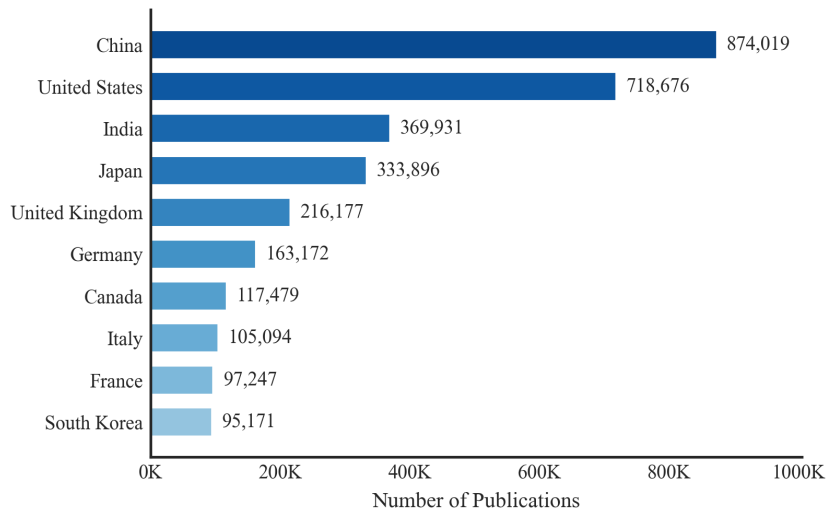


Figure 6: Top 10 countries by AI research output (abstract-level corpus, 2013-2026). A single paper with co-authors from multiple countries is counted once per country.

Japan’s position at rank 4 (333,896) in the abstract corpus is notable. Many Japanese research papers in robotics, materials science, and plasma physics discuss neural networks in their methods sections without placing them in their titles, which means abstract-level search captures a larger share of Japanese AI research than title-only approaches would.

3.7 INSTITUTIONAL OUTPUT

Table 5: Top 10 institutions by paper count.

Rank	Institution	Country	Papers (Count)	Share of Corpus
1	Chinese Academy of Sciences	China	74,921	1.50%
2	CNRS (French Natl. Research Centre)	France	50,145	1.00%
3	University of London	UK	34,887	0.70%
4	Tsinghua University	China	30,519	0.61%
5	Univ. of Chinese Academy of Sciences	China	23,911	0.48%
6	Shanghai Jiao Tong University	China	23,695	0.47%
7	Zhejiang University	China	23,246	0.46%
8	Harvard University	US	21,529	0.43%
9	US Department of Energy	US	20,244	0.40%
10	Peking University	China	20,127	0.40%

The Chinese Academy of Sciences leads with 74,921 papers, 49.4% more than CNRS (50,145). Six of the top ten institutions are Chinese. Harvard (21,529) is the highest-ranked US institution at position 8.

Note that OpenAlex’s institution taxonomy includes umbrella organizations (CNRS, Helmholtz Association, US Department of Energy) alongside individual universities. These umbrella organizations aggregate papers from their constituent laboratories and institutes, which inflates their counts relative to standalone universities.

3.8 OPEN ACCESS

Of the 5,003,783 works in the corpus, 3,043,557 (60.8%) are published as open access (OA) literature, while 1,960,226 (39.2%) remain behind publisher paywalls. For context, Piwowar et al. [11] estimated the baseline open access rate across all academic fields at 28% in 2018. The higher rate in this corpus is consistent with the AI community’s preprint culture, where arXiv is a common platform for early dissemination.

3.9 TITLE VS. ABSTRACT COMPARISON

Table 6: Title-only vs. abstract search for selected keywords.

Keyword	Title Search	Abstract Search	Abstract Only	Ratio
neural network	433,556	1,522,612	1,089,056	3.5x
machine learning	495,798	1,287,123	791,325	2.6x
deep learning	374,975	980,070	605,095	2.6x
large language model	71,469	292,873	221,404	4.1x
diffusion model	42,120	324,073	281,953	7.7x
transformer	129,524	316,216	186,692	2.4x
hallucination	10,711	48,759	38,048	4.6x
fairness	78,835	429,288	350,453	5.4x
retrieval augmented	8,137	27,394	19,257	3.4x
federated learning	39,481	59,298	19,817	1.5x
reinforcement learning	104,021	201,098	97,077	1.9x
knowledge graph	28,201	90,234	62,033	3.2x

The ratio of abstract-to-title matches varies from 1.5x (“federated learning”) to 7.7x (“diffusion model”). Methods that are commonly used as tools rather than as the primary topic of a paper have

the highest ratios. “Diffusion model” (7.7x) is discussed in 324,073 abstracts but placed in the title of only 42,120 papers. Many of these papers use diffusion models as a component of a larger system without naming them in the title.

“Fairness” (5.4x) and “hallucination” (4.6x) show high ratios because they are frequently discussed as secondary concerns in a paper’s abstract rather than as the paper’s primary topic. Note that both terms have non-AI uses: “hallucination” also appears in psychiatric and neuroscience literature, and “fairness” in social science and economics. Their abstract counts may include some non-AI papers.

“Federated learning” (1.5x) has the lowest ratio, meaning papers that discuss federated learning almost always include it in their title. This suggests that federated learning is typically the main contribution of the paper, not a supporting technique.

This finding has methodological consequences for bibliometric research. Title-only analysis systematically undercounts methods that are used as tools across disciplines. Abstract-level analysis captures a more complete picture of method adoption.

4 DISCUSSION

4.1 THE PERSISTENCE OF NEURAL NETWORKS

As shown in Section 3.2, “neural network” remains the most frequently referenced AI concept in the corpus, appearing in 30.4% of all abstracts. This dominance persists despite public attention moving to large language models. Two factors underpin it. First, foundational optimization methods such as Adam [4] and standard libraries like PyTorch [20] have stabilized neural network training, making the architecture accessible across disciplines. Second, the 3.5x ratio between abstract and title counts (Table 6) indicates that most papers using neural networks do not place the term in their titles. Neural networks have become a standard tool rather than a novel contribution. The temporal data (Section 3.3) shows steady 8.9x growth over 12 years with no signs of plateauing.

4.2 THE LLM INFLECTION POINT

The LLM trajectory (Section 3.3) shows the sharpest inflection in the corpus. No other method matches this acceleration profile. What the raw growth numbers do not capture is the nature of the surrounding research. The fastest-rising keywords (Table 4) show researchers building systems around LLMs, not just training them. RAG has become the standard pattern for connecting language models to external knowledge. “Hallucination,” “guardrail,” and “jailbreak” indicate growing attention to reliability and safety.

The rapid standardization of “LLM” as an abbreviation (16.0x growth, Table 4) is itself a signal. LLMs have become a recognized category in AI research vocabulary, similar to how “CNN” and “RNN” became standard abbreviations in previous waves.

4.3 METHOD LIFECYCLES

The time-series data in Section 3.3 reveals different methods at different lifecycle stages.

Mature methods (steady growth). “Neural network” and “knowledge graph” show consistent growth without acceleration or deceleration. These methods have large, established research communities.

Growth phase. “Deep learning,” “transformer,” “graph neural,” and “federated learning” are all growing faster than the corpus average.

Plateau candidates. “Generative adversarial” growth has slowed since 2020, as GANs are being supplemented by diffusion models for many image generation tasks.

Continued growth. “Large language model” shows no signs of deceleration. The annualized 2026 estimate (approximately 170,000 papers) suggests continued acceleration over the 2025 count of 96,984. These lifecycle patterns are summarized visually in Figure 7.

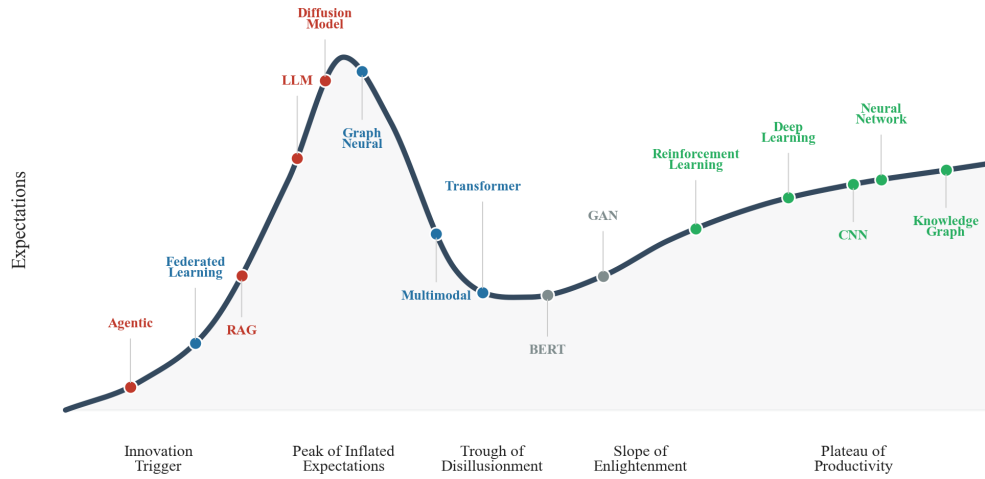


Figure 7: Hype cycle for methods in the corpus. Colors indicate lifecycle category: red (hype peak), blue (growth phase), green (foundational/mature), gray (declining). Placement is interpretive, based on growth trajectories, not a quantitative model.

4.4 THE CHINA-US RESEARCH BALANCE

4.4.1 THE CROSSOVER

China and the United States started the decade at near-parity. In 2013, the US produced 13,829 AI-related papers (by abstract count) while China produced 12,074. Both countries grew steadily through 2018, but their trajectories diverged after that.

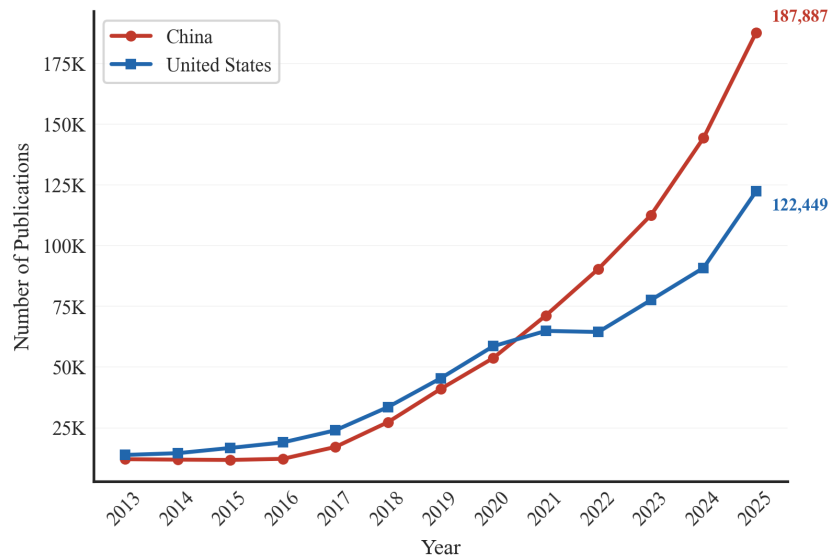


Figure 8: AI research output for China and the United States, 2013-2025. China exceeded US output starting in 2021, with the difference increasing annually.

Table 7: Year-by-year AI research output for top 5 countries.

Year	China	United States	India	Japan	UK	China/US
2013	12,074	13,829	2,761	2,223	4,312	0.87
2014	11,897	14,551	3,445	2,277	4,557	0.82
2015	11,766	16,713	3,976	2,478	5,030	0.70
2016	12,229	18,988	4,996	2,913	5,762	0.64
2017	17,125	23,991	5,919	3,616	7,057	0.71
2018	27,353	33,567	8,592	5,075	9,567	0.81
2019	41,011	45,404	11,585	6,508	12,732	0.90
2020	53,743	58,622	17,612	7,712	16,574	0.92
2021	71,273	64,931	26,158	8,854	19,788	1.10
2022	90,485	64,486	35,760	9,420	19,798	1.40
2023	112,646	77,664	48,990	11,095	24,429	1.45
2024	144,452	90,915	65,398	12,657	28,174	1.59
2025	187,887	122,449	89,287	15,558	37,104	1.53

The crossover occurred in 2021. That year, China produced 71,273 papers while the US produced 64,931. The US output actually declined between 2021 and 2022 (64,931 to 64,486), while China continued to accelerate. By 2025, the gap had widened to 53.4% (187,887 vs. 122,449).

The US deceleration from 2020 to 2022 is notable. US AI research output grew 10.0% over two years (58,622 to 64,486), compared to 68.4% growth for China over the same period (53,743 to 90,485). Growth resumed in the US from 2023 onward (77,664 to 122,449 by 2025, a 57.7% increase over two years), but not fast enough to close the gap.

An important caveat: paper counts measure research volume, not research impact. This study does not analyze citation-weighted metrics, shares of top-1% highly cited papers, or venue prestige, which may yield different rankings. Publication incentive structures also differ across countries. Both China and the US have institutional pressures — tenure requirements, h-index targets, and ranking criteria — that can inflate output independently of research contribution.

4.4.2 METHOD-SPECIFIC COMPARISONS

The China-US balance varies by research area.

Table 8: China vs. US paper counts by method (2013-2026, abstract search).

Method	China	US	Combined	China/US
transformer	89,302	33,506	122,808	2.67x
federated learning	16,356	8,689	25,045	1.88x
neural network	332,858	177,752	510,610	1.87x
deep learning	241,216	140,965	382,181	1.71x
reinforcement learning	50,910	31,277	82,187	1.63x
diffusion model	66,832	56,315	123,147	1.19x
large language model	35,923	47,363	83,286	0.76x

In the corpus, China leads in six of seven method categories. The lead is strongest in “transformer” (2.67x), “federated learning” (1.88x), and “neural network” (1.87x). But the US leads in “large language model” (47,363 vs. 35,923, or 1.32x the Chinese count). This is a meaningful exception. While China produces more AI papers overall in the sample, the US produces more papers on the fastest-growing technology category since 2023.

4.4.3 THE LLM CONVERGENCE

In the corpus, the US led China in LLM research output throughout 2020-2024, with the gap narrowing overall, though it temporarily widened in 2023 before resuming its convergence. In 2020, the US produced 2.4x more LLM papers than China in the corpus. By 2025, China reached parity (China: 15,008 vs. US: 14,735, a ratio of 1.02). This convergence coincides with the release of

Chinese LLMs such as DeepSeek-V3 [7], Qwen, and Yi, which gave Chinese researchers domestic foundation models to study, benchmark, and extend.

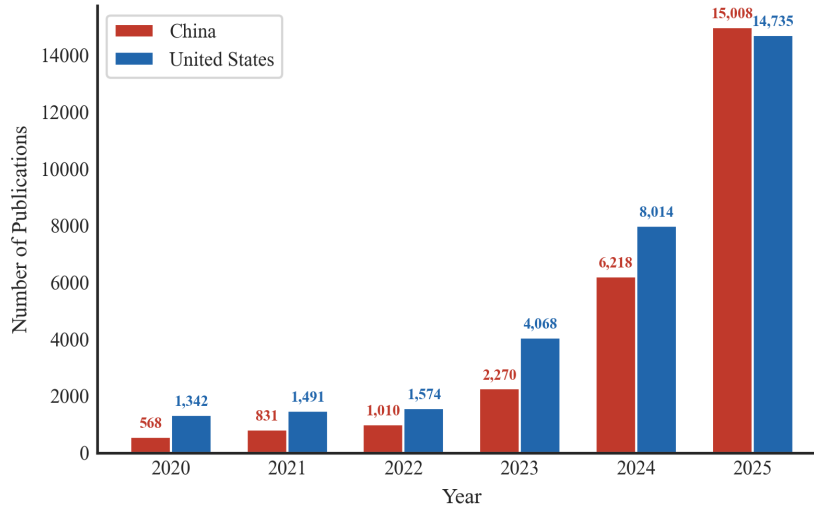


Figure 9: LLM research papers per year for China and the United States. The US led from 2020-2024, with China reaching parity in 2025.

The convergence pattern in the data has three phases. First, the US led comfortably from 2020-2022 as LLM research was concentrated at US-based labs. Second, the gap narrowed in 2023-2024 as Chinese labs released competitive open-weight models. Third, parity was reached in 2025, when Chinese LLM paper output in the corpus matched the US for the first time.

4.4.4 INDIA’S ACCELERATION

India shows a distinct growth pattern. In the corpus, Indian AI research output grew from 2,761 papers in 2013 to 89,287 in 2025, a 32.3x increase. This growth rate is faster than both China (15.6x) and the US (8.9x) over the same period. India produced 89,287 AI papers in 2025, more than Japan (15,558) and the UK (37,104) combined.

India’s growth rate has been accelerating. Between 2013 and 2018, Indian output grew 3.1x. Between 2018 and 2025, it grew 10.4x.

4.5 LIMITATIONS

Stemming and noise. OpenAlex applies stemming to search queries, which inflates counts for common words. “Explainable” returns 2,544,915 abstract matches, which is clearly inflated by stemming matching “explain” in non-AI contexts. Single-word keyword counts in Table 3 are best interpreted with this caveat in mind. Multi-word phrases and proper nouns are minimally affected.

Cross-disciplinary noise. As noted in Section 2.2, some fraction of the corpus consists of non-AI papers matching broad terms like “autonomous” or “generative.” This is an inherent tradeoff of abstract-level analysis versus title-level analysis (which is more precise but less complete).

Temporal coverage. The 2026 cohort covers January through early June. Annualized projections assume even distribution throughout the year, which may not hold due to conference deadlines and journal publication cycles.

Multi-counting. A paper co-authored by researchers in China and the United States is counted once in each country’s total. Country-level paper counts therefore sum to more than the corpus total. The same applies to institutions.

OpenAlex coverage. OpenAlex indexes over 250 million works but does not cover all academic literature. Non-English publications, conference proceedings from smaller venues, and technical reports may be underrepresented. OpenAlex has stronger coverage of English-language journals and repositories.

Causation vs. correlation. Growth in keyword frequency reflects research attention, not research quality or real-world deployment. A 16x increase in “LLM” papers does not mean LLMs are 16x more useful than they were in 2022.

Volume vs. impact. All country and institutional comparisons are based on paper counts, which measure volume, not impact. The caveats discussed in Section 4.4.1 — citation-weighted metrics, venue prestige, and differing publication incentive structures — apply to all geographic and institutional comparisons in this paper.

Cross-disciplinary inclusion. As noted in Section 3.5, the most-cited papers in the corpus include non-AI entries (e.g., DSM-5, lme4) whose abstracts happen to mention AI-related terms. This is an inherent feature of abstract-level search.

No normalization against baseline. The growth rates in this paper are raw, not normalized against overall academic publishing growth. Some of the observed AI growth reflects baseline expansion of scholarly output rather than AI-specific acceleration.

5 RELATED WORK

5.1 AI BIBLIOMETRIC STUDIES

The Stanford HAI AI Index Report [9] is a widely cited annual survey of AI research trends. The 2025 edition tracks publications, patents, investment, and policy across multiple data sources including Dimensions, Epoch, and LMSYS. This study differs in two ways. First, it uses abstract-level search rather than subject classification, capturing cross-disciplinary AI method usage. Second, every count in the analysis can be verified through the free OpenAlex API, rather than requiring access to proprietary databases.

Zhang et al. (2021) conducted a bibliometric analysis of deep learning research using Web of Science data, finding that China and the US together accounted for over 50% of deep learning publications [14]. The abstract-level data in this study is consistent with this finding. China (874,019) and the US (718,676) together account for approximately 32% of all papers in the corpus discussing AI methods, though this percentage is lower because the corpus includes a broader set of documents.

5.2 COMPUTE AND SCALING STUDIES

Sevilla et al. (2022) analyzed compute trends in machine learning, documenting distinct scaling eras with doubling times ranging from 5-6 months in the deep learning era to approximately 10 months in the large-scale era [13]. The publication growth data in this study is consistent with their findings. The periods of fastest publication growth (2017-2018 and 2023-2025) align with the periods when compute scaling enabled new model capabilities.

Hoffmann et al. (2022) introduced compute-optimal scaling laws (“Chinchilla scaling”) [16], and Kaplan et al. (2020) characterized neural scaling laws [17]. These papers provided the theoretical foundation for the compute scaling race. The data on the growth of “large language model” (29.9x from 2018 to 2025) reflects the research activity that this scaling race generated, catalyzed by models like GPT-3 [21].

5.3 RESEARCH COMMERCIALIZATION

Jurowetzki et al. (2021) used arXiv and patent data to map the AI research and development system, finding increasing overlap between academic research and commercial development [15]. The observation that named models (DeepSeek, Claude, Gemini, Mistral, LLaMA) are the fastest-growing terms in the research literature supports this finding. Academic researchers increasingly study commercial AI products, and commercial labs increasingly publish in academic venues.

Ahmed and Wahed (2020) examined the growing compute divide between well-funded industry labs and academic institutions, documenting how resource asymmetry shapes what research can be conducted and by whom [18]. While the data in this study does not directly measure researcher movement, the growth of model-specific research keywords is consistent with a field where commercial products are the objects of study.

5.4 METHODOLOGICAL CONTRIBUTIONS

This study contributes a methodological point to the bibliometrics literature. The title vs. abstract comparison (Table 6) quantifies the information loss in title-only bibliometric analyses. The finding that abstract search captures 1.5x to 7.7x more papers per keyword matters for researchers designing bibliometric studies, confirming standard recommendations for thorough search strategies in literature synthesis [22].

6 CONCLUSION

Five million works, analyzed through abstract-level keyword search, reveal an AI research field shaped by three concurrent trends. Established methods (neural networks, deep learning, reinforcement learning) continue to dominate by accumulated volume. The LLM category has grown faster than other methods in this corpus (29.9x over seven years). And a growing body of research on reliability and safety (hallucination, guardrail, jailbreak) indicates increasing attention to the practical challenges of deploying these systems.

Six principal findings stand out.

1. **Neural networks remain dominant.** “Neural network” appears in 1,522,612 paper abstracts (30.4% of the corpus). This dominance has not diminished despite the attention given to LLMs.
2. **LLMs are the fastest-growing category.** “Large language model” grew from 3,248 abstracts in 2018 to 96,984 in 2025 (29.9x), with an inflection point at the release of ChatGPT.
3. **The field is expanding from architecture toward application.** The fastest-rising terms are not architectures but patterns (RAG, 52.4x), safety concepts (jailbreak, 25.5x), and specific model names (DeepSeek, 848.7x).
4. **China leads in volume.** In the corpus, China produces 21.6% more AI research papers than the United States by paper count. This study does not measure citation impact or venue prestige, which may yield different rankings.
5. **Half of all papers go uncited.** 48.9% of papers have zero citations, indicating extreme concentration of academic impact. This figure is inflated by recent publications that have not yet had time to accumulate citations.
6. **Title-only analysis misses most AI research.** Abstract search captures 1.5x to 7.7x more papers per keyword, depending on the method. Studies that rely on title-level filtering provide an incomplete view of AI research activity.

REFERENCES

- [1] OpenAlex. "OpenAlex API Documentation." <https://docs.openalex.org>
- [2] K. He, X. Zhang, S. Ren, and J. Sun. "Deep Residual Learning for Image Recognition." CVPR, 2016. <https://arxiv.org/abs/1512.03385>
- [3] O. Ronneberger, P. Fischer, and T. Brox. "U-Net: Convolutional Networks for Biomedical Image Segmentation." MICCAI, 2015. <https://arxiv.org/abs/1505.04597>
- [4] D. P. Kingma and J. Ba. "Adam: A Method for Stochastic Optimization." ICLR, 2015. <https://arxiv.org/abs/1412.6980>
- [5] Y. LeCun, Y. Bengio, and G. Hinton. "Deep Learning." Nature, vol. 521, pp. 436-444, 2015. <https://doi.org/10.1038/nature14539>
- [6] A. Vaswani et al. "Attention Is All You Need." NeurIPS, 2017. <https://arxiv.org/abs/1706.03762>
- [7] DeepSeek-AI. "DeepSeek-V3 Technical Report." 2024. <https://arxiv.org/abs/2412.19437>
- [8] P. Lewis et al. "Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks." NeurIPS, 2020. <https://arxiv.org/abs/2005.11401>
- [9] Stanford HAI. "AI Index Report 2025." <https://aiindex.stanford.edu>
- [10] I. Goodfellow et al. "Generative Adversarial Nets." NeurIPS, 2014. <https://arxiv.org/abs/1406.2661>
- [11] H. Piwowar et al. "The State of OA: A Large-Scale Analysis of the Prevalence and Impact of Open Access Articles." PeerJ, 2018. <https://doi.org/10.7717/peerj.4375>
- [12] B. McMahan et al. "Communication-Efficient Learning of Deep Networks from Decentralized Data." AISTATS, 2017. <https://arxiv.org/abs/1602.05629>
- [13] J. Sevilla et al. "Compute Trends Across Three Eras of Machine Learning." 2022. <https://arxiv.org/abs/2202.05924>
- [14] D. Zhang et al. "The AI Index 2021 Annual Report." Stanford HAI, 2021. <https://aiindex.stanford.edu/report/>
- [15] R. Jurowetzki et al. "The Privatization of AI Research(-ers): Causes and Potential Consequences." 2021. <https://arxiv.org/abs/2102.01648>
- [16] J. Hoffmann et al. "Training Compute-Optimal Large Language Models." NeurIPS, 2022. <https://arxiv.org/abs/2203.15556>
- [17] J. Kaplan et al. "Scaling Laws for Neural Language Models." 2020. <https://arxiv.org/abs/2001.08361>
- [18] N. Ahmed and M. Wahed. "The De-democratization of AI: Deep Learning and the Compute Divide in Artificial Intelligence Research." 2023. <https://arxiv.org/abs/2010.15581>
- [19] J. Priem, H. Piwowar, and R. Orr. "OpenAlex: A fully-open index of scholarly works, authors, venues, institutions, and concepts." arXiv preprint arXiv:2205.01833, 2022. <https://arxiv.org/abs/2205.01833>
- [20] A. Paszke et al. "PyTorch: An Imperative Style, High-Performance Deep Learning Library." NeurIPS, 2019. <https://arxiv.org/abs/1912.01703>
- [21] T. B. Brown et al. "Language Models are Few-Shot Learners." NeurIPS, 2020. <https://arxiv.org/abs/2005.14165>
- [22] C. Lefebvre et al. "Searching for studies." Cochrane Handbook for Systematic Reviews of Interventions, 2019. <https://doi.org/10.1002/9781119536604.ch4>
- [23] American Psychiatric Association. "Diagnostic and Statistical Manual of Mental Disorders (DSM-5)." 5th ed., 2013. <https://doi.org/10.1176/appi.books.9780890425596>
- [24] D. Bates, M. Mächler, B. Bolker, and S. Walker. "Fitting Linear Mixed-Effects Models Using lme4." Journal of Statistical Software, vol. 67, no. 1, 2015. <https://doi.org/10.18637/jss.v067.i01>